SBMS UI SCREEN AND WORKFLOW INVENTORY

Prepared: 2026-11-02 **Revision described:** `fix/sbms-completion-work` @ `d6b5032` **Sources read:** `frontend/src/App.jsx`, every file in `frontend/src/pages/`, `frontend/src/components/`, `frontend/src/context/AuthContext.jsx`, `frontend/src/hooks/useHashToken.js`, `frontend/src/services/api.js`, and the three browser harnesses in `scripts/`.

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1. Application shell and routing

The frontend is a single-page React application built with Vite and routed with `react-router-dom`. Every route is declared in `frontend/src/App.jsx`; there is no server-side rendering and no separate admin application.

Path	Component	Guard	Roles admitted
`/login`	`Login`	none	anyone
`/admin/login`	`Login role="admin"`	none	anyone (endpoint decides)
`/field-officer/login`	`Login role="field_officer"`	none	anyone (endpoint decides)
`/verify`	`Verification`	none	public
`/verify-certificate`	`CertificateVerification`	none	public
`/dashboard`	`DashboardHome`	authenticated	admin renders the dashboard; officer is redirected to `/field-officer`
`/field-officer`	`FieldOfficerDashboard`	authenticated	field officer
`/students`	`Students`	`ProtectedPage`	field officer
`/students/register`	`StudentRegistration`	`ProtectedPage`	field officer
`/eligibility`	`Eligibility`	`ProtectedPage`	field officer
`/duplicate-review`	`DuplicateReview`	`ProtectedPage`	admin
`/disbursements`	`Disbursements`	`ProtectedPage`	admin and field officer
`/reports`	`Reports`	`ProtectedPage`	admin
`/settings`	`Settings`	`ProtectedPage`	admin
`/audit`	`AuditLogs`	`ProtectedPage`	admin
`/users`	`UserManagement`	`ProtectedPage`	admin
`/`	redirect	authenticated	to `/dashboard`

`ProtectedRoute` waits for `authLoading`, redirects an unauthenticated visitor to `/login`, and redirects a role mismatch to `/dashboard`. The guard is a **convenience, not a control**: every page's data still arrives from role-guarded endpoints, so a user who forced a route into the DOM would see an empty page and 403 responses rather than another role's data.

2. Shared frontend infrastructure

Element	File	Behaviour
Axios instance	`services/api.js`	`baseURL` is `VITE_API_ORIGIN + /api` when configured, otherwise `/api`; a request interceptor attaches `Authorization: Bearer <token>` from `localStorage`
Auth context	`context/AuthContext.jsx`	Loads the current user from `GET /auth/me` on mount, exposes `isAuthenticated`, `user`, `authLoading`, sign-in and sign-out
Layout	`components/AppLayout.jsx`	Sidebar navigation with role-specific entries, header with the signed-in user and a sign-out action
Public layout	`components/PublicVerificationLayout.jsx`	Minimal centred shell used by both public verification pages, with no navigation and no authenticated data
Hash-token hook	`hooks/useHashToken.js`	Reads `window.location.hash`, captures the token during render (so React StrictMode's double effect cannot strip it before the second pass), removes it with `history.replaceState`, and re-runs on `hashchange`
Certificate panel	`components/AwardCertificatePanel.jsx`	Issue, read, print and revoke controls for an award certificate; used inside the disbursement screen

The hash-token hook exists because of a real defect: verifying once on mount left a stale result when a second QR code was opened in the same tab. That defect, its fix and the StrictMode regression it exposed are recorded in `docs/qr-verification-2026-10-30.md` §4.

3. Public screens

3.1 `/verify` — student QR verification

Property	Detail
Data loaded	None on mount; a token from the fragment is POSTed to `/api/verify`
Success state	A single line, "Record found"
Refusal states	A single line for an unknown, malformed or absent token; "Verification is temporarily unavailable" for a 429 or 503
Data shown	**Nothing else.** No name, school, date of birth, identifier or phone number
URL hygiene	The token is removed from the visible address bar before the request is sent
Evidence	`scripts/qr-smoke.mjs` 9/9 (2026-10-30) asserts each of the above, including that a query-string token is rejected and that a second code in the same tab re-verifies

3.2 `/verify-certificate` — certificate authenticity

Property	Detail
Data loaded	Token from the fragment POSTed to `/api/certificates/verify`
Success state	Authenticity facts: certificate reference, award period, issue date, approval date, issuing authority
Refusal states	`not_recognised`, `revoked`, `not_current`, or a temporary-unavailability line
Data shown	No student name, UPI, school or amount
Evidence	API contracts covered by `certificateVerificationRoute.test.js` (5) and `bursaryCertificates.test.js` (12); **no browser harness exists for this page**

3.3 Sign-in screens

Route	Behaviour
`/login`	Shared sign-in that presents both roles
`/admin/login`	Administrator sign-in; posts to `/api/auth/login/admin`
`/field-officer/login`	Field-officer sign-in; posts to `/api/auth/login/field-officer`

A wrong-role attempt returns 403 from the API with a plain sentence telling the user which door to use, and the page surfaces that sentence rather than a generic failure.

4. Field-officer screens

4.1 `/field-officer` — officer workspace

Property	Detail
Purpose	Entry point for a field officer after sign-in; shows the officer's next actions rather than administrator totals
Data loaded	Uses the authenticated user from context; no administrator endpoint is called
Actions	Navigation to registration, registry and eligibility

4.2 `/students` — registry list

Property	Detail
Data loaded	`GET /students`
Shown	Student rows returned by the endpoint
Actions	Open a student (single read), navigate to registration
Limitation	The endpoint has no pagination or filter parameters, so the screen renders the whole array as returned; a large register would be rendered client-side

4.3 `/students/register` — registration and screening confirmation

Property	Detail
Data loaded	`GET /students/schools` for the school picker
Inputs	Full name (required), date of birth, school (required, five options), national ID, primary phone, guardian ID, **UPI**, **birth certificate number**
Validation	Client-side required fields; the server-side Zod schema is authoritative and its `errors` block is surfaced
Success state	The created student, the QR image, and plain-language screening information: a possible-match sentence, the exact-identifier sentence when applicable, and counts ("checked against N existing students; M flagged; K recorded in the audit log without a separate case")
Refusal state	The API's error sentence in an error panel, with **no confirmation panel**
Deliberate omission	No model version, no cut-off, no score and no `risk_level`: the officer sees wording, not model internals
Evidence	`scripts/browser-smoke.mjs` checks 2-14; `scripts/registration-fields-smoke.mjs` 11/11 against the live deployment

The UPI and birth-certificate inputs were added on 2026-11-01 because the critical exact-identifier rule could otherwise fire only for seeded rows. Both are optional: a guardian may not have the document at the counter, and a blank box is dropped from the payload rather than sent as an empty string.

4.4 `/eligibility` — rule-based scoring

Property	Detail
Data loaded	`GET /students` for the student picker
Inputs	Fee balance, household income, vulnerability status, siblings in school, prior bursary history
Action	`POST /eligibility`
Result shown	Score, category, per-factor breakdown and the reason sentences produced by `services/eligibility.js`
Persistence	Score, category and assessment time are written back to the student row
Limitation	The arithmetic has no automated test (FR-12), so a defect in the banding would not be caught by the suite

4.5 `/disbursements` — awards, payments, reconciliation and certificates

Property	Detail
Data loaded	`GET /awards`, `GET /students`
Award actions	Submit application (`POST /awards`), record verification (`POST /awards/:id/verification`), approve or reject (`PATCH /awards/:id/decision`)
Payment actions	Record payment (`POST /disbursements`), change payment status (`PATCH /disbursements/:id/payment-status`), record acknowledgement (`PATCH /disbursements/:id/acknowledgement`), link a legacy payment (`PATCH /disbursements/:id/award`)
Certificate panel	`AwardCertificatePanel` issues, reads, prints and revokes certificates through `/awards/:id/certificate*`
Role behaviour	The page renders for both roles; controls are permitted by the API, so an officer attempting an administrator-only action receives 403
Known asymmetry	The student picker calls `GET /students`, which is officer-only; for an administrator that call returns 403 and the picker has no data (inspection finding, `SBMS_API_ENDPOINT_CATALOGUE.md` §7.2)
Evidence	Payment and award rules are covered by 20 API tests; **no browser harness drives this screen**

5. Administrator screens

5.1 `/dashboard` — administrative dashboard

Property	Detail
Data loaded	`GET /admin/dashboard`
Shown	Registry, duplicate, award and payment totals; a six-month registration and payment trend; recent audit activity
Limitation	Each counter's arithmetic is not individually tested (FR-25); the endpoint is authorised and the screen is browser-checked as reachable

5.2 `/duplicate-review` — adjudication queue

Property	Detail
Data loaded	`GET /duplicate-flags`
Shown	Per case: priority band, model score, model version, cut-off, reason, masked identifiers. Policy line: both cut-offs, the aggregation policy, and the statement that the thresholds are provisional
Actions	Select a case, reveal identifiers (`GET /duplicate-flags/:id/identifiers`, audited), record a decision (`PATCH /duplicate-flags/:id`) with a reason of at least five characters
Colour rule	Red only for exact identifier agreement; amber for any other review case; green when nothing is flagged
Deliberate design	No control can approve, reject or pay anything; the screen produces a decision record and the award and payment gates read that record
Evidence	`scripts/browser-smoke.mjs` checks 15-23 (queue contents, both cut-offs, model version, "score is not a decision", audit counters)

5.3 `/reports` — PDF export

Property	Detail
Data loaded	`GET /reports/export?start_date=&end_date=`
Actions	Choose a range and download the PDF
Validation	The API refuses a missing range and a reversed range; the screen surfaces the message
Limitation	The PDF's content is verified by inspection only (FR-26)

5.4 `/settings` — duplicate cut-offs

Property	Detail
Data loaded	`GET /settings/threshold`

Property	Detail
Inputs	High-priority cut-off, review cut-off, model review cut-off
Action	`PUT /settings/threshold`
Validation	Each value 0-1; the high-priority cut-off must be greater than or equal to the review cut-off
Persistence	Process memory only; a restart restores environment defaults (`SBMS_ML_SUBSYSTEM_DOCUMENTATION.md` §5)

5.5 `/audit` and `/users`

Screen	Data loaded	Actions	Notes
`/audit`	`GET /audit-logs`	Read; screening counters are rendered as readable text rather than raw JSON	Append-only by convention only
`/users`	`GET /users`	Create an account (`POST /users`), change a role (`PATCH /users/:id/role`)	Self-demotion and last-administrator removal are refused by the API

6. Workflow walkthroughs as the interface presents them

Workflow	Screens in order	What the user sees at each step
Register and screen a beneficiary	`/field-officer/login` → `/students/register` → confirmation panel	Sign-in → form → created student, QR image, plain-language screening outcome
Adjudicate a duplicate	`/admin/login` → `/duplicate-review` → case detail → decision form	Queue with priority → masked identifiers, score, cut-offs → reason prompt → confirmation that the decision was recorded
Award and pay	`/students/register` → `/disbursements` (submit, verify, approve) → record payment → clear → acknowledge	Each step either succeeds or returns a named reason (not verified, duplicate open, amount exceeded, evidence missing)
Identify a student	Printed QR → phone camera → `/verify`	One line: record found or not, and nothing else
Verify a certificate	Printed certificate → `/verify-certificate`	Authenticity facts, or revoked / not current / not recognised
Administrative review	`/dashboard`, `/reports`, `/audit`, `/users`, `/settings`	Totals, exported PDF, history, accounts, cut-offs

7. Interface states

State	Presentation
Loading	`ProtectedRoute` renders nothing while `authLoading` is true, so a protected page cannot flash before authentication resolves
Empty	Lists render with no rows rather than a placeholder message
Error	Panels show the server's `message`; a 5xx shows `Internal server error`, never a stack
Refused action	The sentence names the missing precondition, e.g. `Verify the beneficiary before approving this award.`
Role mismatch	Redirect to `/dashboard`, or a 403 sentence naming the correct sign-in page

8. Accessibility and responsiveness: what is and is not established

Aspect	State
Layout	Responsive, single-column on narrow viewports
Keyboard operation	Standard form controls and buttons; no custom widget with bespoke key handling was introduced
Colour	Status colour is always accompanied by text, so meaning does not depend on colour alone
Contrast, font scaling, screen-reader labelling	**Not audited**
Automated accessibility testing	**None**

Aspect	State
Usability testing with officers	**None** (G-06)

NFR-14 must therefore be reported as partially verified on the strength of browser checks that assert rendered text, classes and navigation, not on any accessibility or usability evidence.

9. Screens that do not exist

Absent screen	Consequence
Password reset or change	A user cannot recover an account unaided, and the live administrator password has already drifted from the stored secret
Student edit or delete	The API exposes create, list and read only; a typo cannot be corrected through the interface
QR token re-issue	A student row without a token cannot be given one (FR-35)
Payment execution	Payments are recorded, never transferred (FR-37)
Case assignment	Deliberately absent: any administrator may take any case
Notification or messaging	No email, SMS or in-app notification exists

10. Mapping to the Chapter Four interface figures

Figure in Chapter Four	Screen	Evidence available for the figure
Fig. 4.13 sign-in	`/login`, `/admin/login`, `/field-officer/login`	Screenshot not captured (G-04)
Fig. 4.14 registry list	`/students`	Screenshot not captured
Fig. 4.15 eligibility	`/eligibility`	Screenshot not captured
Fig. 4.16 registration form	`/students/register`	Screenshot not captured; field list verified by the 11-check harness
Fig. 4.17 screening confirmation	`/students/register` success panel	Screenshot not captured; text verified by browser checks 3-7
Fig. 4.18 dashboard	`/dashboard`	Screenshot not captured
Fig. 4.19 duplicate review	`/duplicate-review`	Screenshot not captured; contents verified by browser checks 15-21
Fig. 4.20 awards and certificates	`/disbursements` plus the certificate panel	Screenshot not captured
Fig. 4.21 administrative support screens	`/reports`, `/audit`, `/users`, `/settings`	Screenshot not captured
Fig. 4.22 student verification page	`/verify`	Screenshot not captured; behaviour verified by 9 browser checks
Fig. 4.23 certificate verification page	`/verify-certificate`	Screenshot not captured; behaviour verified at API level only

Every screenshot must be taken against synthetic data and checked for real beneficiary details before insertion (G-05). No screenshot in this folder may show a live deployment record created during testing, including the two `E2E Probe` students named in the deployment log.